

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Supplementary Summer Examination – 2025****Course: T.Y. BTECH.****Branch: Mechanical Engineering****Semester: V****Subject Code & Name: BTMEC504\_19 Theory of Machines- II****Max Marks: 60****Date: 24-07-2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                                                                                                                                            | (Level/CO) | Marks     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                                                                                                                                                                     |            | <b>12</b> |
| 1           | <b>The effect of increasing the diameter of the driving pulley in a belt drive system is:</b><br>a. Increase the speed of the driven pulley    b. Decrease the speed of the driven pulley    c. Increase the power transmission    d. No effect on the driven pulley speed |            | 1         |
| 2           | <b>In a V-belt drive, the angle of the V-shaped groove helps to:</b><br>a. Increase the tension in the belt    b. Reduce slipping    c. Increase the diameter of the pulleys    d. Decrease the efficiency                                                                 |            | 1         |
| 3           | <b>The module is the reciprocal of.....</b><br>a. Diametral pitch    b. circular pitch    c. pitch diameter    d. none of the mentioned                                                                                                                                    |            | 1         |
| 4           | <b>The size of a gear is usually specified by</b><br>a. pressure angle    b. circular pitch    c. Diametral pitch    d. pitch circle diameter                                                                                                                              |            | 1         |
| 5           | <b>When the axes of first and last gear are co-axial, then gear train is known as</b><br>a. simple gear train    b. compound gear train    c. reverted gear train    d. Epicyclic gear train                                                                               |            | 1         |
| 6           | <b>The type of gears used to connect two non-parallel and non-intersecting shafts is</b><br>a. Spur gear    b. Helical gear    c. Bevel gear    d. Spiral gear                                                                                                             |            | 1         |
| 7           | <b>A governor is said to be hunting, if the speed of the engine</b><br>a. remains constant at the mean speed    b. is above the mean speed    c. is below the mean speed    d. fluctuates continuously above and below the mean speed                                      |            | 1         |
| 8           | <b>The axis of rotation of the gyroscope's spinning wheel is called the _____</b>                                                                                                                                                                                          |            | 1         |

|      |                                                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  |    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------|------------------------------------|--|----|
|      | a. Precession axis                                                                                                                                                                                                                                                                                                     | b. Input axis                   | c. Spin axis                              | d. Output axis                     |  |    |
| 9    | In aeroplanes, gyroscopic couple is most significant during:                                                                                                                                                                                                                                                           |                                 |                                           |                                    |  | 1  |
|      | a. Straight flight                                                                                                                                                                                                                                                                                                     | b. Banking                      | c. Rapid pitch changes (take-off/landing) | d. Engine shutdown                 |  |    |
| 10   | For occurrence of free torsional vibration which of the condition is necessary?                                                                                                                                                                                                                                        |                                 |                                           |                                    |  | 1  |
|      | a. Rotors moving in same direction                                                                                                                                                                                                                                                                                     | b. Rotors having same frequency | c. Rotors having different frequency      | d. Rotors rotate in the same sense |  |    |
| 11   | The factor which affects the critical speed of a shaft is                                                                                                                                                                                                                                                              |                                 |                                           |                                    |  | 1  |
|      | a. Diameter of the disc                                                                                                                                                                                                                                                                                                | b. Span of the shaft            | c. Eccentricity                           | d. All of the mentioned            |  |    |
| 12   | The speed at which the shaft runs so that the additional deflection from the axis of rotation of the shaft becomes infinite, is known as                                                                                                                                                                               |                                 |                                           |                                    |  | 1  |
|      | a. Whirling speed                                                                                                                                                                                                                                                                                                      | b. Rotational speed             | c. Stabilizing speed                      | d. Reciprocating speed             |  |    |
|      |                                                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  |    |
| Q. 2 | Solve the following.                                                                                                                                                                                                                                                                                                   |                                 |                                           |                                    |  | 12 |
| A)   | A flat belt drive transmits 35 kW from a pulley of 1.5 m effective diameter running at 300 rpm. The angle of contact is spread over 11/24 of the circumference, and the coefficient of friction between the belt and pulley surface is 0.3. Determine the width of the belt required, considering centrifugal tension. |                                 |                                           |                                    |  | 6  |
| B)   | Why are rope drives suitable for long-distance power transmission? Explain with reference to their efficiency and design.                                                                                                                                                                                              |                                 |                                           |                                    |  | 6  |
|      |                                                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  |    |
| Q.3  | Solve the following.                                                                                                                                                                                                                                                                                                   |                                 |                                           |                                    |  | 12 |
| A)   | Explain the concept of involute gear tooth profile. Why is it preferred over cycloidal profiles?                                                                                                                                                                                                                       |                                 |                                           |                                    |  | 6  |
| B)   | A pinion with 25 teeth drives a gear with 75 teeth. The pinion rotates at 1200 rpm and transmits 200 Nm torque. Find:<br>a) Speed of the gear<br>b) Torque on the gear<br>c) Power transmitted by the gear pair                                                                                                        |                                 |                                           |                                    |  | 6  |
|      |                                                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  |    |
| Q. 4 | Solve Any Two of the following.                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  | 12 |
| A)   | Describe a compound gear train. How does it improve velocity ratio in limited space?                                                                                                                                                                                                                                   |                                 |                                           |                                    |  | 6  |
| B)   | A flywheel of moment of inertia 50 kg·m <sup>2</sup> runs at 200 rpm. If the speed fluctuates between 180 rpm and 220 rpm during the cycle, find the energy stored in the flywheel.                                                                                                                                    |                                 |                                           |                                    |  | 6  |
| C)   | Explain the working of a differential gear using a simple epicyclic gear train. Why is it used in vehicles?                                                                                                                                                                                                            |                                 |                                           |                                    |  | 6  |
|      |                                                                                                                                                                                                                                                                                                                        |                                 |                                           |                                    |  |    |

|             |                                                                                                                                                                                                                           |  |           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                    |  | <b>12</b> |
| <b>A)</b>   | Define and discuss the terms: sensitivity, stability, hunting, and isochronism in governors.                                                                                                                              |  | <b>6</b>  |
| <b>B)</b>   | Discuss the function of a flywheel and how it differs from that of a governor.                                                                                                                                            |  | <b>6</b>  |
| <b>C)</b>   | A gyroscope has a rotor of mass 5 kg and radius of gyration 0.2 m, rotating at 3000 rpm. The ship in which it is mounted turns with an angular velocity of 0.1 rad/s. Calculate the gyroscopic couple acting on the ship. |  | <b>6</b>  |
|             |                                                                                                                                                                                                                           |  |           |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                    |  | <b>12</b> |
| <b>A)</b>   | Discuss the effects of gyroscopic couple on ships, aeroplanes, and vehicles, with examples.                                                                                                                               |  | <b>6</b>  |
| <b>B)</b>   | A mass of 10 kg is attached to a spring with stiffness $k=2000\text{N/m}$ . Determine: a) Natural frequency in Hz b) Natural angular frequency in rad/s.                                                                  |  | <b>6</b>  |
| <b>C)</b>   | Explain the concept of torsional vibration in rotating shafts. Discuss its causes and effects on mechanical systems.                                                                                                      |  | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                                                        |  |           |